YULIM (JESS) CHOI

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EDUCATION

University of Southern California, Viterbi School of Engineering — **Los Angeles, CA**
Master of Science in Applied Data Science — Aug. 2026 – May 2028

Sungkyunkwan University (SKKU) — **Seoul, Korea**
Bachelor of Library and Information Science; Bachelor of Business Administration — Feb. 2022 – Feb. 2026

- GPA: 3.86/4.00 (Magna Cum Laude)
- Thesis: "From Metadata Gaps to a Reuse-Oriented Data Lifecycle Model: A Quantitative Analysis of the National Library of Korea's Bibliographic LOD Using DCC and FAIR"
- Academic Excellence Scholarship – Top 3% (Highest Honors, Spring 2023); Top 5% (Honors, Fall 2024), College of Liberal Arts

Indiana University, Kelley School of Business — **Bloomington, IN**
Exchange Student — Aug. 2024 – May 2025

TECHNICAL SKILLS

Python (pandas, NumPy, scikit-learn, Matplotlib, Seaborn) | SQL | R

RESEARCH INTERESTS

Applied Machine Learning, Human-Centered AI, Responsible AI, AI Safety

PUBLICATION

IK Choi, M Čukić, **YL Choi,** "ECG-based Autonomic Profiles for Depression: Partial Reproduction, Nonlinear Extension, and Commentary," *Research Square (Sep. 2026, preprint; not peer-reviewed)* [DOI]

RESEARCH EXPERIENCE

SKKU Department of Library and Information Science — **Seoul, Korea**
Undergraduate Research Assistant (Advisor: Prof. Inkyung Choi) — Oct. 2025 – Aug. 2026

FAIR-MIND: Biomarker-Driven Mental Health Data Interoperability

- Reviewed literature on biomarker-based depression assessment, comparing machine learning, data analytics, and ontology frameworks to identify methodological gaps and guide research.
- Drafted key sections of "FAIR-Mind: Addressing Critical Gaps in Mental Health Data Interoperability through Community-Driven Application Profiles" (FAIR Connect, February 2026), synthesizing research on biomarker-based precision psychiatry and examining limitations in semantic interoperability and ontology-based frameworks.
- Developed an end-to-end ECG processing pipeline that selected 72,468 eligible recordings; generated analysis-ready profiles for 35,980 unique subjects through data cleaning, quality control, feature engineering, and aggregation.
- Integrated data across clinical cohorts, resolving feature compatibility issues to standardize downstream analysis.

Building a Materials Science Knowledge Graph Using MatSciBert

- Built a Python preprocessing workflow for 56 full-text materials science papers in XML format, using section-level parsing, text normalization, and sentence segmentation to create a 4,622-sentence corpus for named entity recognition (NER).
- Cleaned and deduplicated the NER corpus, reducing 18,488 text examples to 3,782 unique examples—a 79.5% reduction—before annotation in Doccano.
- Designed a domain-specific NER annotation framework for fluoride-ion battery literature, defining six entity categories and rules for span boundaries, abbreviations, ambiguity resolution, and non-overlapping annotations toward consistent labeling.
- Developed an end-to-end literature-to-annotation pipeline integrating API-based retrieval, XML acquisition, text cleaning, sentence-level chunking, and NER pre-annotation to generate Doccano-ready JSONL files.
- Administered a university-hosted Doccano environment, managing dataset imports and exports, user access, annotation records, and SQLite database backups.

ACADEMIC PROJECTS

Relationships between Loneliness in Older Adults and Technology Use | *Canva*

- Designed infographics visualizing technology adoption trends among older adults in the U.S. and South Korea.
- Analyzed survey data and extracted cross-cultural differences in loneliness, digital engagement, and societal attitudes toward technology shaped by regional characteristics.
- Examined the effects of cultural norms and policy environments on social connectedness among older adults.
- Conducted literature reviews to contextualize findings and support evidence-based recommendations for interventions.

Database Design for Operational Efficiency | *SQL*

- Designed and normalized a relational database for Cerulean Bee to streamline apparel order management.
- Created ER diagrams and translated them into conceptual, logical, and physical schemas to enforce data integrity.
- Implemented the system in Oracle using DDL and DML, developing queries and procedures to support reporting and minimize redundancy.

Analyzing Influential Factors of Traffic Stops in San Francisco | *R*

- Built a decision tree model and binomial simulation to detect demographic/contextual factors affecting traffic stop outcomes.
- Cleaned and transformed 900K+ rows, reducing dataset size by 10% to optimize model efficiency and minimize load times.
- Simulated 10,000 traffic stop scenarios to quantify disparities and assess enforcement risks with statistical precision.

Proposed Direction to Address South Korea's Low Reading Rate | *Python*

- Preprocessed the National Reading Promotion Plan dataset to filter irrelevant records and craft analytical target variables.
- Visualized library usage and sociodemographic correlations using Seaborn and Matplotlib, identifying key factors influencing reading engagement; derived policy implications in alignment with Korea's Fourth National Basic Plan for Reading Culture.

Suicide Rates and Suspicious Factors by Gender, Age, and Region | *Tableau*

- Developed dashboards comparing Korea's suicide rates per 100K with OECD averages, GDP-peer nations (Mexico, Australia), and culturally similar Japan.
- Analyzed disparities by region, gender, and age, highlighting male rates at 2× females and sharp increases after age 70.
- Assessed socioeconomic and mental health indicators alongside suicide prevention policies, revealing limited effectiveness.

Passenger Congestion Prediction at Hyehwa Subway Station| *Python*

- Built predictive models using ML algorithms such as K-Nearest Neighbors, Linear Regression, and Random Forest to forecast daily congestion levels at Hyehwa Station.
- Consolidated fragmented ridership data into a unified dataset, integrating temporal and seasonal features.
- Visualized passenger flow patterns using Seaborn and Matplotlib to uncover peak congestion periods and seasonal variability.

Global Life Expectancy Inequality: Health–Economic Relationship Analysis| *Python*

- Preprocessed data from 179 countries by removing irrelevant columns and restructuring analytical frames.
- Analyzed correlation using NumPy and heatmaps to clarify links between life expectancy, health indices, and economic factors.
- Compared high- vs. low-life-expectancy regions using statistical metrics and visualizations, revealing infant/child mortality, HIV incidence, and GDP per capita as major determinant of inequality.

TEACHING & VOLUNTEERING

Global Summer Program, Brown University – *Student Mentor* Summer 2025

- Coordinated and led cultural immersion activities for 8 visiting Brown U. students, facilitating experiential learning and cross-cultural understanding/communication between the U.S. and Korea.

NSLI-Y Program, U.S. Department of State – *Korean Language Tutor* Jun. 2023 – Jul. 2023

- Tutored American high school students in Korean language fundamentals and organized cultural activities.

Good Neighbors – *Remote Volunteer Translator* Feb. 2023 – Dec. 2024

- Translated 190 global sponsor letters from Korean to English ensuring cultural sensitivity, clarity, and emotional resonance.